AERIS: Adaptive Environment-Aware Routing for IoT Sensors under Heterogeneous WSN Channels

Anonymous Author(s)
Submission for IEEE LCN 2026 (double-blind review)

***Abstract*—AERIS (*Adaptive Environment-aware Routing for IoT Sensors*) is a fixed-rule hierarchical routing protocol for wireless sensor networks (WSNs) under heterogeneous channel conditions. It combines context-adaptive intra-cluster forwarding, Gateway-assisted cluster-head-to-base-station (CH-to-BS) uplinks, and a sparse Skeleton reserve backbone. In NS-3, we compare AERIS against LEACH, PEGASIS, HEED, TEEN, and simplified CTP/RPL-MRHOF baselines across 50–1000 nodes and four environments. The results show a deployment boundary rather than universal superiority: AERIS ranks first in 21 of 28 cells against classical baselines, but its advantage narrows to outdoor suburban settings once collection-style baselines are included. A 3360-run ablation study and a 400-replicate mechanism analysis attribute the gain mainly to Gateway-assisted uplinks, while also revealing a trade-off: earlier first-node death (FND) in large, harsh regimes. AERIS is therefore best read as a reliability-first option for specific harsh environments, not as a general replacement for mature collection-tree stacks.**

***Index Terms*—wireless sensor networks, reliable routing, cross-platform evaluation, deployment boundary**

I. INTRODUCTION

Wireless sensor network (WSN) routing is traditionally evaluated through the lens of energy efficiency, yet in practical deployments, link stability often dictates the fundamental usability of a routing rule [1]–[3]. Classical protocols address this trade-off via distinct architectural priorities: LEACH focuses on clustered forwarding [4], PEGASIS employs chain-based aggregation [5], HEED incorporates residual-energy-aware head election [6], and TEEN restricts reporting to event-triggered thresholds [7]. In real-world scenarios, the problem is compounded because low-power WSN channels are inherently asymmetric, environment-sensitive, and prone to instability near connectivity thresholds [8]–[10].

The practical challenge is not to identify a single dominant routing family. It is to find the regimes where a simple, auditable controller outperforms classical energy-first designs, and the regimes where richer collection-tree or RPL-style stacks should remain the default. AERIS addresses that question with fixed-rule control rather than learned heuristics. Its architecture combines Context-Adaptive Switching (CAS) for intra-cluster mode selection, Gateway reinforcement for CH-to-BS uplinks, and a sparse Skeleton reserve backbone.

We make three primary contributions:

1) We present AERIS as a fixed-rule routing design whose online decisions are traceable to channel and topology inputs.
2) We evaluate it with a five-protocol NS-3 comparison and a seven-protocol boundary sweep, including simplified CTP/RPL-MRHOF baselines and a collision-aware stress layer, while keeping ranking evidence separate from stress evidence.
3) Through ablation and pooled trade-off statistics, we show that the main reliability gain comes from Gateway support and is accompanied by earlier node death.

II. RELATED WORK

Classical low-overhead WSN protocols continue to define the primary engineering operating points. LEACH and HEED mitigate long-range transmission costs via clustering, though their uplink quality remains highly sensitive to head placement and channel degradation [6], [11]. PEGASIS distributes load through chain forwarding, but this robustness incurs the cost of increased latency and serialization [5]. While TEEN suppresses redundant transmissions, its logic makes performance metrics sensitive to the specific definition of "expected" delivery [7]. Collectively, these protocols represent the reliability-energy-latency trade-off that any contemporary design must confront.

Recent research has sought to improve reliability through machine learning, hybrid control, and context-aware adaptation [12]–[18]. AERIS aligns with the objective of context-awareness but deliberately maintains a simple runtime logic: it relies on fixed coefficients and explicit control paths without training-time dependencies. Alongside a broader trend toward complex AI-driven routing and sophisticated LLN management documented in recent surveys [19]–[21], emerging designs frequently explore reinforcement learning and federated DDQN strategies [22]–[25].

Collection-tree and RPL-style protocols remain critical because they address the parent-selection problem through structured control planes. CTP constructs trees based on path quality [26], while RPL and MRHOF standardize DAG construction for lossy networks [27], [28]. Opportunistic variants like ORW and ORPL further exploit link diversity [29], [30]. This paper does not seek to replace these mature families. Instead, it investigates where a simpler fixed-rule design provides a sufficient reliability gain and where the established collection-tree paradigm should remain the default choice.

III. METHOD

AERIS integrates CAS for intra-cluster mode selection, Gateway reinforcement for cluster-head-to-base-station uplinks, and a sparse Skeleton relay backbone as a reserve

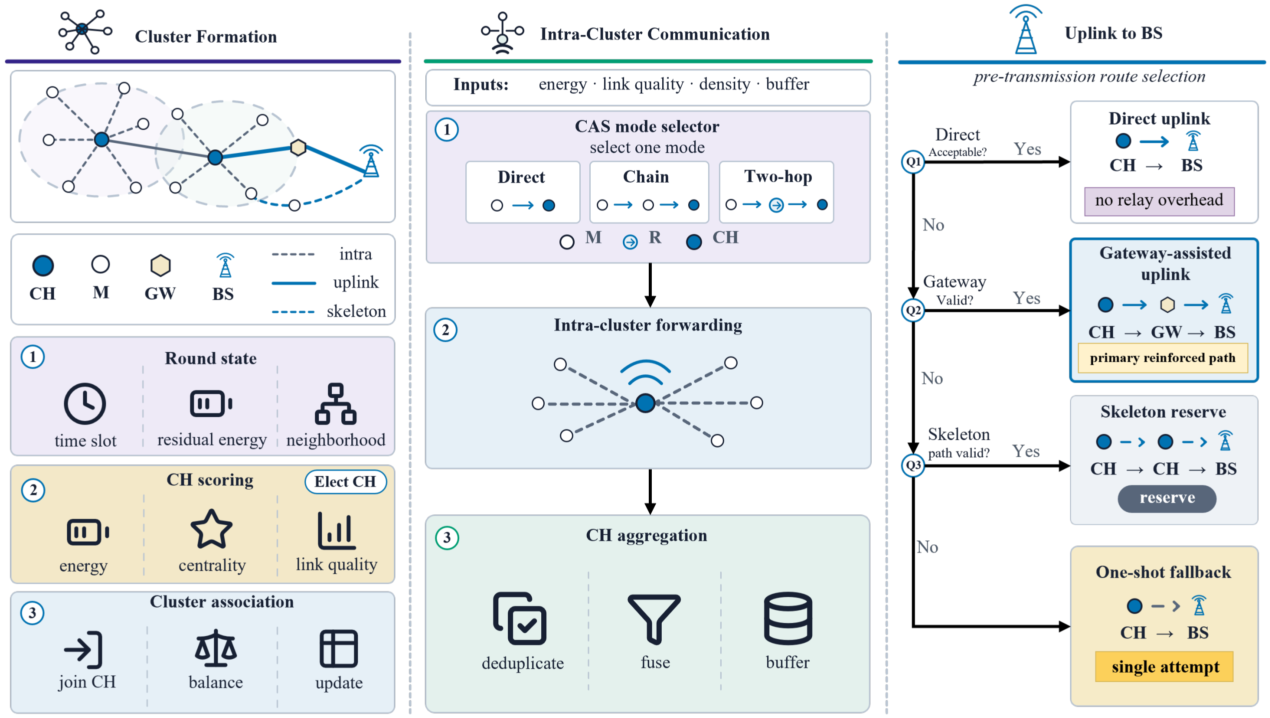

Fig. 1. AERIS round framework. The protocol first forms clusters and scores CH/Gateway candidates, then selects one CAS intra-cluster forwarding mode, and finally applies a first-hit uplink selector over direct, Gateway-assisted, Skeleton-reserve, and one-shot fallback paths.

fallback. The design adds redundancy only where direct forwarding becomes fragile, while keeping the online control path deterministic and auditable. Figure 1 summarizes the framework in the same order: cluster formation, intra-cluster communication, and pre-transmission uplink selection to the BS.

CH candidates are scored by the fuzzy `cluster_head_chance` output of `FuzzyLogicSystem`, which combines residual energy, node centrality, node degree, distance-to-BS, and link quality. The CH fraction starts from 0.08, increases under weak mean LQI, is adjusted for density, and is clipped to $[0.05, 0.20]$. A fairness penalty reduces repeated CH duty by multiplying the fuzzy score by $1 - 0.25p_i$, where p_i is the historical CH-use penalty.

Gateway candidates are ranked by a fixed linear score,

$$\text{Score}_i = w_D\bar{D}_i + w_C\hat{C}_i + w_L\hat{L}_i + w_E\hat{E}_i, \tag{1}$$

where $\bar{D}_i$ is normalized CH-to-BS distance, and $\hat{C}_i$, $\hat{L}_i$, and $\hat{E}_i$ are normalized centrality, link quality, and residual energy. The publication setting uses $w_D = -0.7$, $w_C = 0.3$, $w_L = 0$, and $w_E = 0$, so closer and more central CHs are preferred unless later configurations explicitly enable link or energy terms.

CAS uses fixed linear scores over {direct, chain, two-hop}:

$$\begin{aligned} m^\star = \arg\max_m (& w_E^m\hat{E} + w_L^m\hat{L} + w_B^m\hat{D}_B \\ & + w_R^m\hat{R} + w_\rho^m\hat{\rho} + w_F^m\hat{F}), \end{aligned} \tag{2}$$

where $\hat{E}$, $\hat{L}$, $\hat{D}_B$, $\hat{R}$, $\hat{\rho}$, and $\hat{F}$ denote residual energy, local link quality, distance-to-BS, cluster radius, density, and fairness penalty. Two-hop also receives a tail bonus $0.2\max(0, \hat{T} - 0.6)$. Rule overrides force two-hop when distance or tail length is high and link quality is below 0.55, and force chain only in dense mid-distance clusters.

Fixed settings and execution. All reported runs use one environment-independent configuration fixed before evaluation: CH/join starts at 0.08 and is clipped to $[0.05, 0.20]$, with join weights $0.45/0.40/0.15$ for distance/link/energy and a minimum PRR of 0.20; Gateway uses $w_D/w_C/w_L/w_E = -0.7/0.3/0/0$ with one retry and direct rescue, while Skeleton stays a PCA-axis reserve; CAS uses EMA/confidence $0.2/0.2$, and the direct, chain, and two-hop rows over (E, L, D_B, R, ρ, F) are $(.35, .65, -.25, -.05, .10, -.05)$, $(.30, .40, .20, .20, .20, -.05)$, and $(.20, .25, .50, .15, .05, -.05)$.

IV. EXPERIMENTS

A. Experimental Setup

The primary metric is packet delivery ratio over expected application-layer transmissions,

$$\text{PDR}_{\text{expected}} = \frac{N_{\text{delivered}}}{N_{\text{expected}}}, \tag{3}$$

where N_{expected} counts one packet per alive source node per round even if a protocol suppresses actual transmission. The denominator is kept fixed across protocols so that the reliability comparison is not confounded by protocol-specific reporting logic.

We use four complementary settings. A five-protocol NS-3 comparison evaluates AERIS against LEACH, PEGASIS, HEED, and TEEN across four environments and 100, 500, and 1000 nodes ($n = 30$ per cell). An expanded NS-3

TABLE I. Experimental settings used in the paper.

Layer	Platform	Prot.	Cells	Role
Classical NS-3	NS-3	5	4 × 3	anchor
Boundary sweep	NS-3	7	4 × 7	boundary
Module ablation	NS-3	4 var.	4 × 7	attribution
Stress layer	Python	5 adapted	4 × 6	stress
Mechanism	Python	AERIS	4 × 3	attribution
Frozen block	pooled	5	100-node	trade-off

boundary sweep adds simplified CTP- and RPL-MRHOF-style collection baselines over 50–1000 nodes. In this standalone harness, CTP and RPL-MRHOF are simplified collection-style baselines rather than complete LLN stacks. Our CTP variant follows the link-cost forwarding rule of [26] but omits the 4-bit beacon-driven link estimator, dynamic beacon timer, and datapath validation. Our RPL-MRHOF variant follows the path-cost parent selection of [27], [28] but omits trickle-driven DIO/DAO, full DODAG repair, and rank-based loop avoidance. These simplifications remove parts of the standards-compliant control plane and repair machinery, so the AERIS–baseline results should be read as a controlled collection-style boundary comparison. Full Contiki-NG/TinyOS stacks may shift the boundary in either direction. Opportunistic variants such as ORW/ORPL [29], [30] and TSCH-based stacks [31] are deferred to future work.

A 3360-run NS-3 ablation compares full AERIS with no-Gateway, no-CAS, and no-CH-score variants. A collision-aware Python stress layer adds explicit collision and relay contention; LEACH, HEED, and TEEN are minimally relay-enabled there for multi-hop comparability. A separate fixed 100-node publication block reports pooled PDR, consumed energy, lifetime, first-node death, and hop count, while a 400-replicate-per-cell mechanism study explains which AERIS mechanisms actually fire.

The NS-3 evidence uses a standalone ns-3.40 validation harness. Nodes are uniformly placed in a 200 m × 200 m field with the sink at the top-center boundary. Runs use 300 rounds, 512-byte packets, 2.0 J initial energy, 10 dBm transmit power, and seeds 42001–42030. The channel combines log-distance loss, shadowing, fading, receiver sensitivity, and packet-error probability. The path-loss/shadowing pairs are office (2.0, 4.5 dB), factory (2.7, 8.5 dB), suburban (2.8, 7.5 dB), and urban (3.4, 12.0 dB). The harness operates above an idealized contention-free MAC abstraction rather than the full IEEE 802.15.4 stack [32]. The collision-aware Python stress layer adds explicit collision and relay contention on top of this abstraction.

The five-protocol NS-3 comparison anchors the classical-baseline ranking; the seven-protocol sweep maps the boundary against simplified collection-style baselines; the ablation attributes the gain; and the collision-aware stress layer provides stress evidence rather than a second canonical ranking. Lifetime, FND, hops, and energy expose load concentration. In Fig. 4, dots use Welch-type cell tests with Hedges' g and Holm correction. In Fig. 2, gaps below 0.1 percentage point are treated as practical near-ties, and winner labels alone should not be read as significance claims.

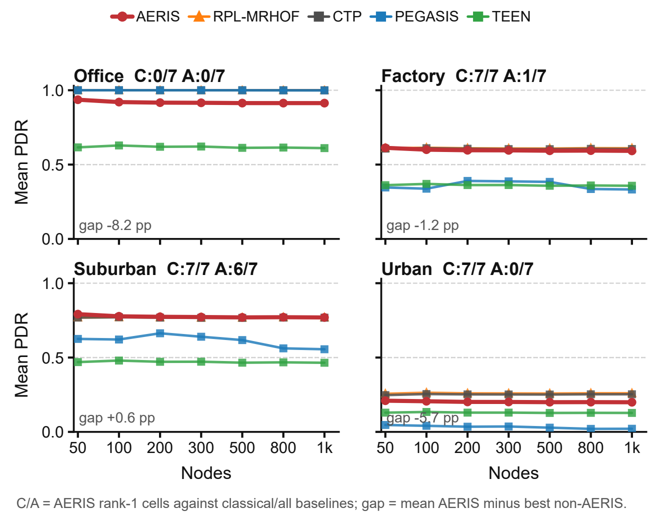

Fig. 2. Expanded NS-3 boundary sweep across four environments and 50–1000 nodes ($n = 30$ per cell). Each panel plots mean PDR for AERIS and the strongest boundary-relevant baselines. The title reports AERIS rank-1 coverage against classical/all baselines (C/A), and the panel note reports the mean signed gap to the best non-AERIS baseline. Gaps below 0.1 percentage point are near-ties.

B. Expanded Boundary Sweep

Figure 2 gives the main boundary result from the seven-protocol NS-3 sweep as PDR trajectories over node scale. Against classical WSN protocols alone, AERIS is rank-1 in 21 of 28 environment-node cells and top-2 in all 28. This supports the narrower classical-baseline claim that AERIS is a reliability-first design that remains effective outside the benign office regime.

The expanded sweep also shows why that claim must be bounded. Once the simplified CTP- and RPL-MRHOF-style baselines are included, AERIS is strongest mainly in outdoor suburban cells: it wins 6 of 7 scales and is essentially tied at 1000 nodes, where the simplified RPL-MRHOF-style baseline leads by only 0.01 percentage points. These cells should be read as near-ties rather than practical superiority claims. In contrast, the CTP-style baseline leads the office cells, and the RPL-MRHOF-style baseline leads most factory and urban cells. AERIS is slightly ahead at factory-50, but falls 1.3–1.8 percentage points behind the RPL-MRHOF-style baseline at larger factory scales; in urban, the average gap to the winner is 5.69 points.

This boundary is the main practical outcome of the expanded sweep. AERIS mitigates the fragility of clustering and chain routing by protecting uplinks with Gateway support, but collection-tree and RPL-style baselines already cover part of the same path-selection problem with richer parent choice.

The strongest defensible claim is therefore not that AERIS is globally best, but that its rule-based control path delivers a large gain over classical WSN baselines and remains competitive or best in selected harsh-channel regimes, especially suburban cells.

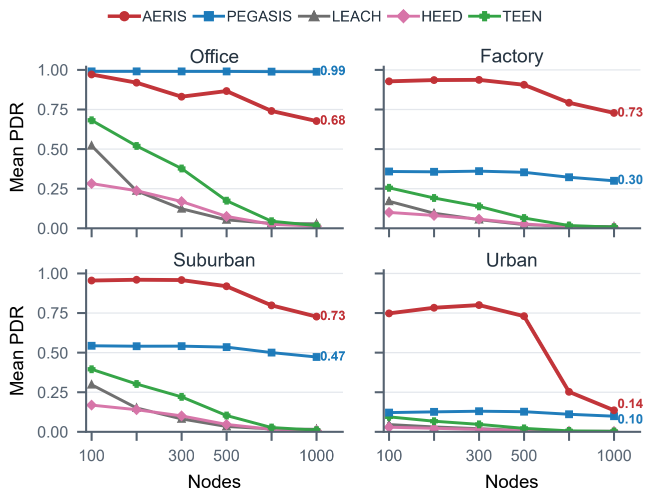

Fig. 3. Collision-aware adapted Python stress test across four environments and 100–1000 nodes ($n = 3200$ per cell). LEACH, HEED, and TEEN are minimally relay-enabled; this layer tests collision-and-relay stress and is not canonical ranking evidence.

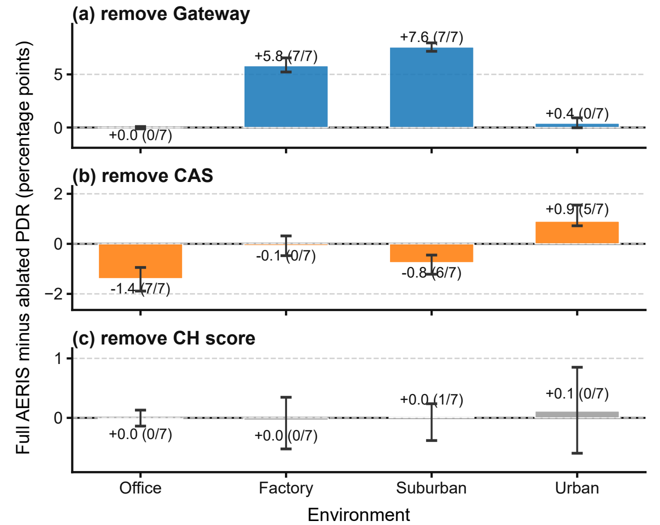

Fig. 4. Expanded NS-3 AERIS ablation across four environments and 50–1000 nodes ($n = 30$ per cell). Each panel summarizes one ablated module as the mean full-minus-ablated PDR contribution by environment; whiskers show the observed node-scale range, and the counts in parentheses give the number of significant node-scale cells after Welch tests with Holm correction.

C. Classical Comparison and Stress Evidence

The five-protocol NS-3 comparison remains the classical-baseline anchor. In that subset, PEGASIS is best in indoor office, while AERIS leads in factory, suburban, and urban conditions. At 1000 nodes, AERIS reaches 0.593 in indoor factory, 0.770 in outdoor suburban, and 0.200 in outdoor urban, compared with 0.324, 0.549, and 0.019 for PEGASIS. The expanded result should therefore be read as a boundary refinement rather than a reversal of the classical-baseline claim.

The collision-aware stress layer asks whether the classical-baseline ordering survives once collision and relay contention are made explicit. Figure 3 suggests that the ordering is largely preserved under this adapted stress model. PEGASIS remains strongest in indoor office (0.991 at 100 nodes and 0.988 at 1000), whereas AERIS stays rank-1 in indoor factory, outdoor suburban, and outdoor urban. At 1000 nodes, AERIS remains well above PEGASIS in indoor factory (0.728 vs. 0.300) and outdoor suburban (0.728 vs. 0.473), and it still leads in outdoor urban (0.136 vs. 0.099).

Because LEACH, HEED, and TEEN are minimally augmented with a simple relay layer for multi-hop comparability in this setting, Fig. 3 is stress evidence rather than a second canonical ranking. Its narrower role is to show that AERIS's harsh-channel advantage survives explicit collision-and-relay stress, even though absolute PDR falls for every protocol in the harder scales.

D. Ablation and Deployment Boundary

Figure 4 summarizes the expanded NS-3 module attribution test. It compares full AERIS with three ablated variants over the four-environment, seven-scale matrix used in the boundary sweep, for 3360 experiments in total. The result is sharper than the earlier fixed 100-node ablation: Gateway is the main gain carrier in factory and suburban cells. Removing Gateway costs 5.82 percentage points on average in factory and 7.58 points in suburban, with all seven node scales significant after Holm correction in both environments. It is essentially neutral in office and weak in urban, which matches the boundary result: office does not need extra uplink protection, while urban links can become too poor for a nearby Gateway to rescue reliably.

CAS has a more mixed role. Removing CAS is beneficial in office and suburban cells, but hurts urban delivery by 0.91 points on average, with five of seven urban scales significant. CAS is therefore not a universal gain module; it is most useful when a short local detour remains viable under harsh links. Removing the CH scoring component is near-neutral in all environments. That does not make the score irrelevant to the design, but it does show that the measured delivery gain in this publication configuration should not be attributed to head scoring.

E. Mechanism and Deployment Trade-off

The 400-replicate-per-cell mechanism study is summarized in Fig. 5. Gateway uplink PDR stays near 1.0 in office, factory, and suburban cells, but drops sharply at urban-1000, exactly where end-to-end PDR collapses. A targeted rerun of the three harsh 1000-node cells on an independent machine reproduced the same bottleneck pattern to the reported precision. The visual takeaway matches the ablation: Gateway carries the gain, CAS is conditional, and Skeleton remains dormant because Gateway support resolves most eligible fallback cases in this configuration. These cell-level FND values expose harsh-regime role exhaustion and are not directly comparable to the pooled 100-node block in Table II.

The trade-off is also clear. Table II shows that AERIS records the lowest consumed energy (131.1 J), but this value reflects early forwarding-role exhaustion rather than an iso-

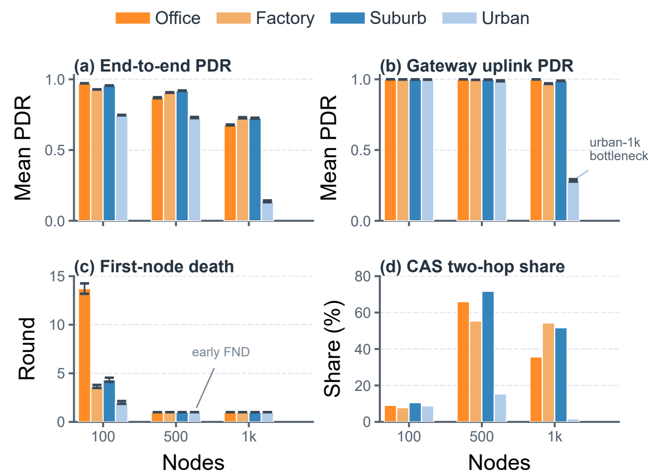

Fig. 5. AERIS mechanism study from 400 replicates per environment-node cell across 12 cells (4800 runs total). The four panels use grouped bars to show end-to-end PDR, Gateway-uplink PDR, first-node death (FND), and CAS two-hop share by environment and node scale.

TABLE II. Separate fixed 100-node trade-off summary. The broader sweeps in Figs. 2–4 cover 50–1000 nodes; this table pools the fixed 100-node block for energy, lifetime, FND, and hop reporting. Bold indicates AERIS. NR = no first-node death before the 300-round horizon.

Method	PDR↑	Consumed J↓	Lifetime↑	FND↑	Hops↓
LEACH	0.260	165.4	300.0	NR	1.71
HEED	0.414	165.2	300.0	11.4	2.00
TEEN	0.432	159.4	299.0	NR	1.28
PEGASIS	0.373	137.6	300.0	70.7	32.37
AERIS	**0.674**	**131.1**	**215.5**	**15.8**	**1.98**

lated efficiency advantage. It also has the shortest lifetime (215.5 rounds), confirming that delivery is obtained through concentrated relay burden. Fig. 5 localizes the cost to early node death and shrinking two-hop support in harsh cells.

Figures 2–4 and Table II suggest a concrete deployment rule. Prefer PEGASIS-like routing when lifetime dominates in benign cells. Consider mature CTP/RPL-style stacks first when richer collection-tree control is acceptable, especially in office, factory, or urban regimes. Choose AERIS when delivery under heterogeneous harsh links is the priority and a simple auditable controller is required, especially in suburban-like cells. Office-like regimes, longevity-first deployments, and mature LLN stacks are the clearest negative cases.

V. LIMITATIONS

Several limits remain. The boundary sweep includes only simplified CTP- and RPL-MRHOF-style collection baselines, not opportunistic routing, ORW/ORPL-style candidate forwarding, coding-based reliability, channel hopping, or standards-compliant Contiki-NG/TinyOS RPL stacks. The AERIS–baseline ordering should therefore be read as a controlled collection-style comparison, not a direct claim against full LLN stacks. The collision-aware stress layer is deployment-stress evidence rather than real-radio-testbed ground truth, and its MAC abstraction omits full 802.15.4/TSCH/CSMA-CA behavior.

Because TEEN is event-triggered, $\mathrm{PDR}_{\mathrm{expected}}$ should be read together with energy, lifetime, FND, and hops; actual-transmission PDR or event-detection success would be needed for a TEEN-centered study. Topology is static, and the environment taxonomy is bounded to the four tested regimes. Recent LLN work likewise emphasizes scalable path management and control-plane flexibility [19], [20]. The provenance chain separates native-harness evidence from stress evidence, and a targeted replication of the three harsh 1000-node mechanism cells reproduced the key bottleneck means. All AERIS coefficients and thresholds are fixed once before evaluation; the implementation and raw results will be released publicly upon acceptance.

VI. CONCLUSION

This paper evaluates AERIS through a five-protocol NS-3 comparison, a seven-protocol boundary sweep, a collision-aware stress layer, and a mechanism study. The claim is deliberately bounded: AERIS is a reliability-first alternative to classical WSN baselines, but richer collection-tree routing narrows or removes that advantage in office, factory, and urban regimes. Its main contribution is a visible deployment boundary, not a universal ranking. Future work should compare against full RPL/ORW/ORPL-style stacks and more realistic MAC behavior, and test whether Gateway-style reinforcement can coexist with collection-tree parent selection without reproducing the early-node-death cost.

REFERENCES

[1] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, and E. Cayirci, "A survey on sensor networks," *IEEE Communications magazine*, vol. 40, no. 8, pp. 102–114, 2002.

[2] T. Rault, A. Bouabdallah, and Y. Challal, "Energy efficiency in wireless sensor networks: A top-down survey," *Computer networks*, vol. 67, pp. 104–122, 2014.

[3] D. Kandris, P. Tsioumas, A. Tzes, G. Nikolakopoulos, and D. D. Vergados, "Power conservation through energy efficient routing in wireless sensor networks," *Sensors*, vol. 9, no. 9, pp. 7320–7342, 2009.

[4] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-efficient communication protocol for wireless microsensor networks," in *Proceedings of the 33rd annual Hawaii international conference on system sciences*. IEEE, 2000, pp. 10–pp.

[5] S. Lindsey and C. S. Raghavendra, "Pegasis: Power-efficient gathering in sensor information systems," in *Proceedings, IEEE aerospace conference*, vol. 3. IEEE, 2002, pp. 3–3.

[6] O. Younis and S. Fahmy, "Heed: a hybrid, energy-efficient, distributed clustering approach for ad hoc sensor networks," *IEEE Transactions on mobile computing*, vol. 3, no. 4, pp. 366–379, 2004.

[7] A. Manjeshwar and D. P. Agrawal, "Teen: Arouting protocol for enhanced efficiency in wireless sensor networks." in *ipdps*, vol. 1, no. 2001, 2001, p. 189.

[8] M. Zuniga and B. Krishnamachari, "Analyzing the transitional region in low power wireless links," in *2004 First Annual IEEE Communications Society Conference on Sensor and Ad Hoc Communications and Networks, 2004. IEEE SECON 2004*. IEEE, 2004, pp. 517–526.

[9] M. Z. Zamalloa and B. Krishnamachari, "An analysis of unreliability and asymmetry in low-power wireless links," *ACM Transactions on Sensor Networks (TOSN)*, vol. 3, no. 2, pp. 7–es, 2007.

[10] N. Baccour, A. Koubâa, L. Mottola, M. A. Zúñiga, H. Youssef, C. A. Boano, and M. Alves, "Radio link quality estimation in wireless sensor networks: A survey," *ACM Transactions on Sensor Networks (TOSN)*, vol. 8, no. 4, pp. 1–33, 2012.

[11] W. B. Heinzelman, A. P. Chandrakasan, and H. Balakrishnan, "An application-specific protocol architecture for wireless microsensor networks," *IEEE Transactions on wireless communications*, vol. 1, no. 4, pp. 660–670, 2002.

[12] D. P. Kumar, T. Amgoth, and C. S. R. Annavarapu, "Machine learning algorithms for wireless sensor networks: A survey," *Information Fusion*, vol. 49, pp. 1–25, 2019.

[13] L. Hu, C. Han, X. Wang, H. Zhu, and J. Ouyang, "Security enhancement for deep reinforcement learning-based strategy in energy-efficient wireless sensor networks," *Sensors*, vol. 24, no. 6, p. 1993, 2024.

[14] G. Pushpa, R. A. Babu, S. Subashree, and S. Senthilkumar, "Optimizing coverage in wireless sensor networks using deep reinforcement learning with graph neural networks," *Scientific Reports*, vol. 15, no. 1, p. 16681, 2025.

[15] F. Tossa, Y. Faga, W. Abdou, E. C. Ezin, and P. Gouton, "Wireless sensor network deployment: Architecture, objectives, and methodologies," *Sensors*, vol. 25, no. 11, p. 3442, 2025.

[16] M. Wang, Z. Zhu, Y. Wang, and S. Xie, "Energy-efficient and highly reliable geographic routing based on link detection and node collaborative scheduling in wsn," *Sensors*, vol. 24, no. 11, p. 3263, 2024.

[17] A. M. Khedr, O. Alfawaz, P. R. PV, and W. Osamy, "Advancing iot-driven wsns with context-aware routing: A comprehensive review," *Computer Science Review*, vol. 58, p. 100803, 2025.

[18] H. Almutairi, S. AlJanah, and N. Zhang, "A secured swarm intelligence-based path selection framework for malicious low-power and lossy networks under rpl protocol," *Internet of Things*, p. 101776, 2025.

[19] H. Almutairi and N. Zhang, "A survey on routing solutions for low-power and lossy networks: Toward a reliable path-finding approach," *Network*, vol. 4, no. 1, pp. 1–32, 2024.

[20] L. Djennadi, G. Diaz, K. Boussetta, and C. Cerin, "Exploring sdn architectures for iot over low-power and lossy networks (llns): A survey," *Computer Networks*, p. 111494, 2025.

[21] S. Thakur, N. I. Sarkar, and S. Yongchareon, "Ai-driven energy-efficient routing in iot-based wireless sensor networks: A comprehensive review," *Sensors*, vol. 25, no. 24, p. 7408, 2025.

[22] J. Ren, J. Zheng, X. Guo, T. Song, X. Wang, S. Wang, and W. Zhang, "Mefi: Mean field reinforcement learning for cooperative routing in wireless sensor network," *IEEE Internet of Things Journal*, vol. 11, no. 1, pp. 995–1011, 2023.

[23] A. A. Okine, N. Adam, F. Naeem, and G. Kaddoum, "Multi-agent deep reinforcement learning for packet routing in tactical mobile sensor networks," *IEEE Transactions on Network and Service Management*, vol. 21, no. 2, pp. 2155–2169, 2024.

[24] P. Singh and R. Vir, "Enhanced energy-aware routing protocol with mobile sink optimization for wireless sensor networks," *Computer Networks*, vol. 261, p. 111100, 2025.

[25] S. N. Bhukya and C. S. R. Annavarapu, "Hybrid reliable clustering algorithm with heterogeneous traffic routing for wireless sensor networks," *Sensors*, vol. 25, no. 3, p. 864, 2025.

[26] O. Gnawali, R. Fonseca, K. Jamieson, D. Moss, and P. Levis, "Collection tree protocol," in *Proceedings of the 7th ACM conference on embedded networked sensor systems*, 2009, pp. 1–14.

[27] T. Tsvetkov, A. Klein *et al.*, "Rpl: Ipv6 routing protocol for low power and lossy networks," *Network*, vol. 59, pp. 59–66, 2011.

[28] O. Gnawali and P. Levis, "The minimum rank with hysteresis objective function," IETF, Tech. Rep. RFC 6719, 2012.

[29] O. Landsiedel, E. Ghadimi, S. Duquennoy, and M. Johansson, "Low power, low delay: Opportunistic routing meets duty cycling," in *Proceedings of the 11th international conference on Information Processing in Sensor Networks*, 2012, pp. 185–196.

[30] S. Duquennoy, O. Landsiedel, and T. Voigt, "Let the tree bloom: Scalable opportunistic routing with orpl," in *Proceedings of the 11th ACM Conference on Embedded Networked Sensor Systems*, 2013, pp. 1–14.

[31] S. Duquennoy, B. Al Nahas, O. Landsiedel, and T. Watteyne, "Orchestra: Robust mesh networks through autonomously scheduled tsch," in *Proceedings of the 13th ACM conference on embedded networked sensor systems*, 2015, pp. 337–350.

[32] P. B. Compatibility, "Ieee standard for low-rate wireless networks," in *IEEE standard for low-rate wireless networks*, 2023.